
Zero Views Museum: Counterfunctional Design for Art Algorithms

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Composite image showing a participant inside *Zero Views Museum's* VR space, surrounded by user-uploaded artworks to Flickr with zero views.

Abstract

Zero Views Museum is a virtual reality experience exhibiting artworks that have never been viewed online - each erased after being seen by a single visitor. This counterfunctional design materializes the invisibility imposed by automated curation systems. Observations revealed emotional responses to ephemeral art and reflection on algorithmic bias. Technical challenges included headset ergonomics and spatial boundaries. By leveraging VR's capacity for solitary experience, this work reframes algorithmic limitations as opportunities for critical artistic inquiry.

Introduction

Our consumption of digital art is increasingly mediated by recommendation systems that prioritize popularity; the more views an artwork has, the better it must be and the more it deserves to be shown.

Zero Views Museum responds to this context by employing counterfunctional design principles [1] to create an immersive virtual reality (VR) experience that exclusively exhibits digital artworks with zero views. Once viewed, works are permanently removed from the database, making participants both the first and potentially last person to see them.

This encounter invites reflection from participants: Are algorithms good arbiters of artistic worth? What unseen works have already been lost? And does our perception of art change when we know we can never see it again?

Context



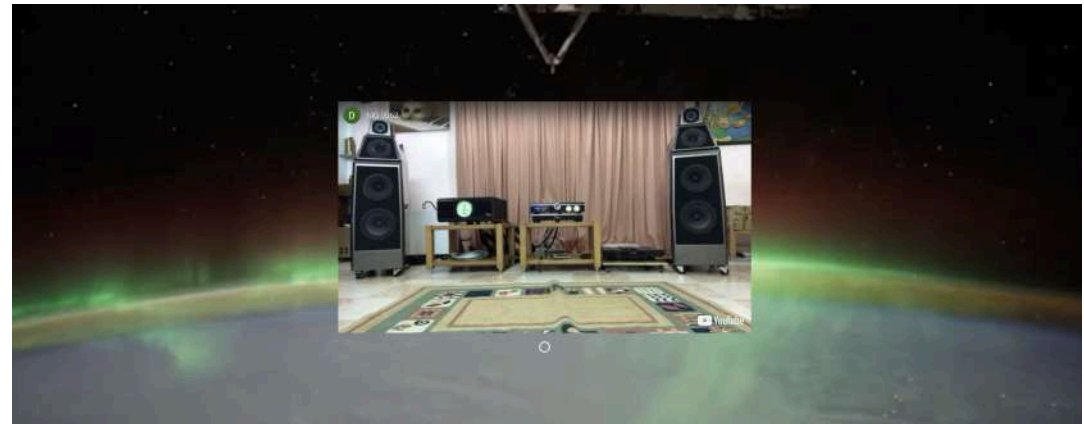
We build on Pierce and Paulos' seminal work in counterfunctional design [1], which explores technological limitations as a design strategy. The inaccessible camera - a digital device encased in concrete - prevented image review without destruction of the camera.



Elsden et al. [2] extends counterfunctionality to Zoom videoconferencing through plugins, such as emoticon "face masks" (above) or theatric webcam filters.



Herman and Moruzzi [3] contrasted human- and Instagram-led art curation through two sides of a physical exhibit, revealing algorithmic preference for instantly recognizable objects over artistic interpretation.



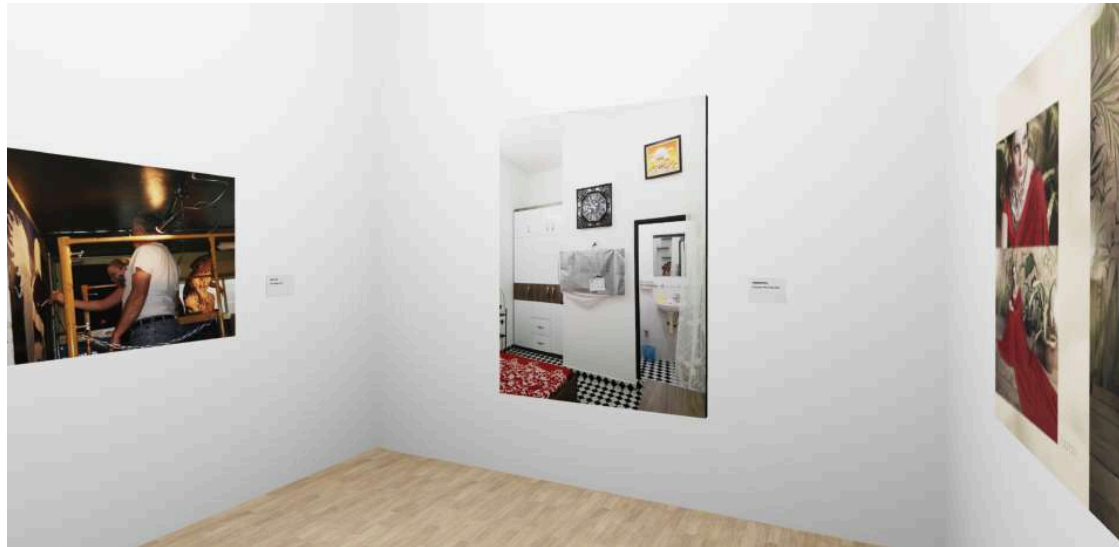
Astronaut.io [4] presents a timed scroll of minimally-viewed YouTube videos overlaid on an astronaut's perspective from space to represent moments of humanity passing by. Other digital experiences also exhibit videos, songs, and games which have been ignored by algorithms [5, 6]. To our best knowledge, we are the first to create an exhibit within VR.

Experience

Only one participant can interact with *Zero Views Museum* using VR at a time. During this, an audience can view a live feed of their movements in a parallel, abstracted space. After each session, the system is manually reset before the next participant begins. [Link to video demo of the experience.](#)



1. Puts on VR headset (Quest 2), adjusting it to their head.



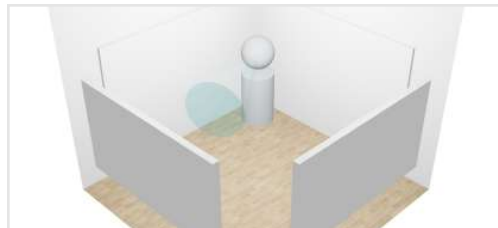
2. Enters VR experience by crossing a 4×4m boundary, chosen to match VR room.

3. Inside VR, freely moves and browses four artworks queried from Flickr with zero views.

4. Exits VR experience by removing VR headset.



5. New artworks are loaded with an "A" button press on the VR controller.



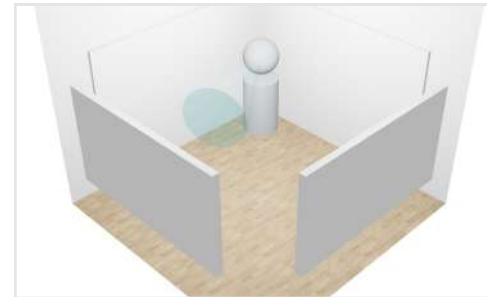
During the experience, an audience can view a live feed of the participant moving and looking around a similar room.

New participant

Concept

We chose the medium of VR to create a unique, individualized, immersive experience for each person. Inside VR, a 1:1 scale of locomotion was used instead of teleportation to prioritize familiarity and leave the participant's hands controller-free.

The audience display was created to draw bystanders into the experience. To represent the anonymity of online browsing and the algorithmic ignorance of artworks, it substitutes the artworks for blank canvases and represents the player as a grey figure reminiscent of social media avatars.



Audience display, featuring blank canvases and a grey avatar figure.



The avatar figure was styled after Facebook's default avatar.

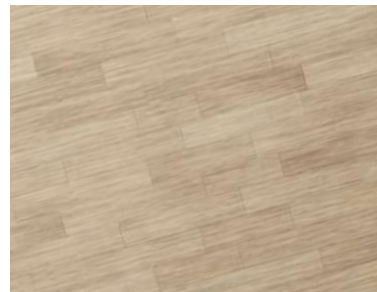


Participant interacting with *Zero Views Museum* inside VR. The audience display can be seen on a monitor on the left, synchronized to the participant's locomotion.

Concept

Inspired by Marcel Duchamp's *Fountain* [7], we similarly wanted to elevate these unseen artworks to the aesthetic standard of an art museum. Thus, the display of artworks, design of descriptive panels, and flooring of *Zero Views Museum* was inspired by New York City's Museum of Modern Art.

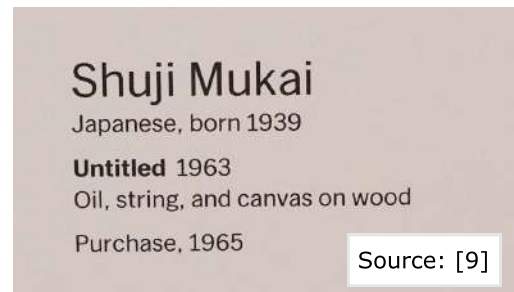
Zero Views Museum, Online



Museum of Modern Art, New York City, USA



Source: [8]



Implementation

Data

We query user-uploaded artworks from Flickr's API [11] using Python's asyncio library [12] and filter to those with zero views. Example search queries included "photography" and "digital art".

```
for query in SEARCH_QUERIES:
    ...
    images = await search_flickr_images(...)
    ...
    if error_code == 429: # Timeout error
        time.sleep(3600) # Wait an hour
        break
    ...
    await save_to_supabase(images, query)
```

Async Requests

Query and date range to search

Artworks with zero views (if any)

```
params = {
    "method": "flickr.photos.search",
    "text": query,
    "media": "photos",
    "per_page": 500,
    "extras": "views,description,...",
    "min_upload_date": min_upload_date,
    "max_upload_date": max_upload_date,
    "safe_search": 1
}
...
zero_view_images = [img for img in images if
int(img.get("views", 1)) == 0]
```

Flickr API

Backend

We store unseen artworks in Supabase [13]. Key fields include the artwork's creator, title, description, and URL for later credit and access. A viewed boolean field is used to filter out displayed artworks.

Artworks with zero views

Artworks with zero views

creator_name	text	title	text	url	text	viewed	bool
DYS Photo III		IMG_8563		https://live.stati		TRUE	
Andika Pradana		Light Painting pa		https://live.stati		FALSE	
Vatimbora Viagens		IMG_1232		https://live.stati		FALSE	
Yuri D		ремеди		https://live.stati		TRUE	
DYS Photo III		IMG_1800		https://live.stati		FALSE	
Cory Barton		DSCN3795		https://live.stati		FALSE	
Don Oddy		DonInLondon Ph		https://live.stati		FALSE	
Jeff Frank		IMG_9672		https://live.stati		FALSE	
Tom Vargas		860420_484397		https://live.stati		FALSE	
		0233		https://live.stati		TRUE	
		antur		https://live.stati		FALSE	

Artwork database

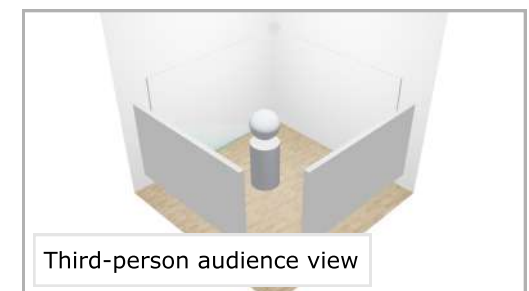
Frontend

We fetch unseen artworks from Supabase to display in a VR museum created with react-three-fiber [14] and pmndrs/xr [15]. Locomotion for the audience display is synchronized with Supabase Realtime [16].



Changes to viewed when displayed

Camera position and rotation





Participant interacting with Zero Views Museum at the CS 889 course showcase.

Discussion

Our demo at the CS 889 course showcase offered valuable insights into how people engage with algorithmically hidden artworks in VR.

Key Takeaways and Implications

The VR medium proved highly effective in conveying the concept of hidden algorithmic curation. Participants were struck by the realism of the museum and the personal nature of the experience—knowing that no one else would see their version of the gallery added emotional weight. For artists and curators, this opens possibilities for exhibitions that play with personalized or ephemeral content, where the medium itself reinforces the message.

The fact that many participants read the descriptive plaques and verbally narrated their experiences—especially when encountering strange or striking artworks—indicates that UX designers should consider how to encourage social sharing even in solitary VR experiences. Additionally, the ease of locomotion (no learning curve for movement) shows that 1:1 walking in VR is highly intuitive across demographics, supporting its use in public installations.

Limitations

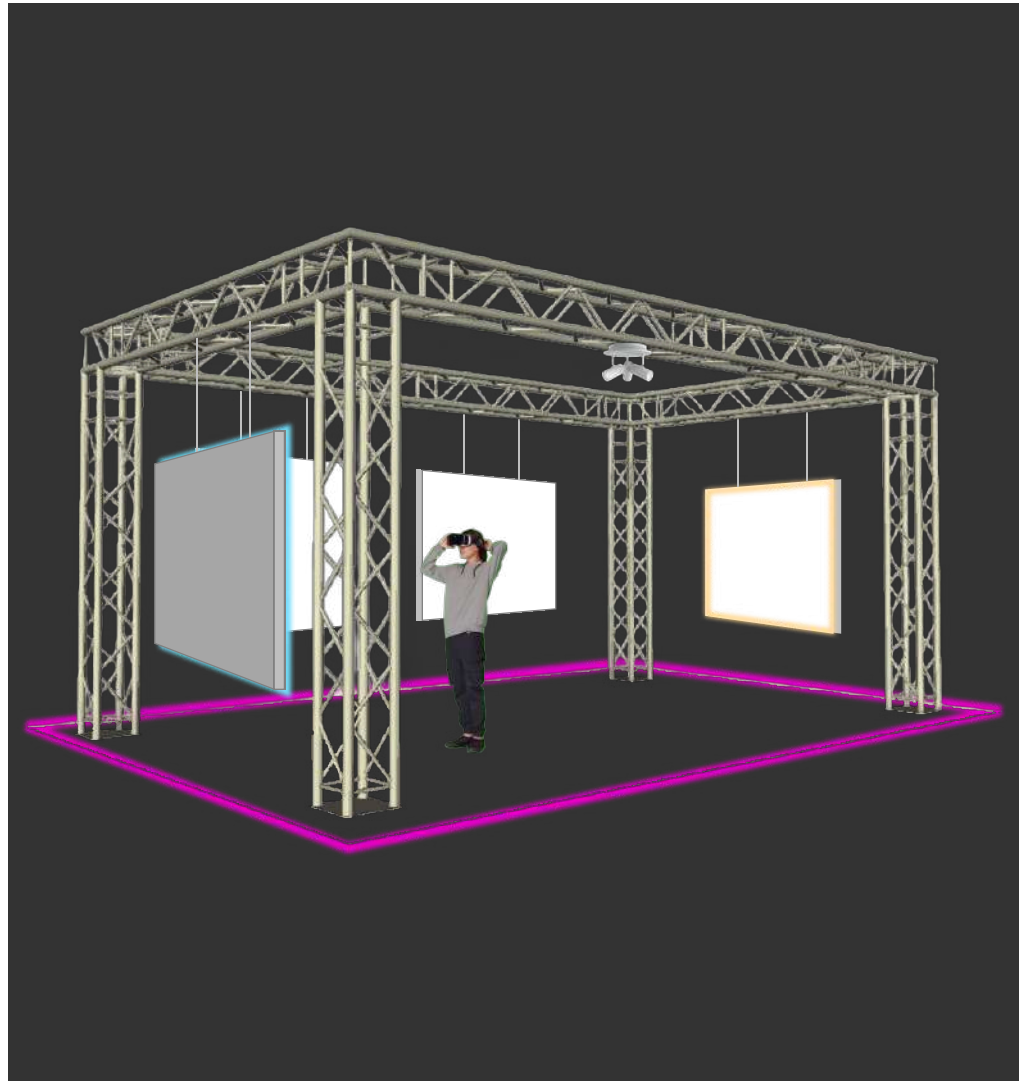
- The Quest 2's headset adjustment was unintuitive for many, leading to discomfort or the need to physically hold the headset while moving. Future iterations should include clearer fitting instructions or consider alternative headsets with easier adjustments.
- Since we intentionally avoided a debug feed to preserve the uniqueness of the experience, troubleshooting display issues was difficult. A non-intrusive monitoring system (e.g., a subtle admin indicator) could help without breaking immersion.
- While tape marked the boundaries, some participants felt uneasy about potential collisions. A more enclosed setup including lightweight barriers could improve perceived safety.

Future Work

This work could evolve in two key directions: First, a hybrid physical-VR installation where the virtual gallery is mapped precisely onto a real-world space containing blank canvases. This stark contrast could make the hidden nature of the content more tangible while also piquing the curiosity of onlookers.

Second, formal studies could compare two versions of the virtual gallery: the current "zero-view" museum against a "popular-only" museum featuring only algorithmically promoted pieces. This comparison could reveal differences in how visitors engage with the content, their emotional responses, and their reflections on curation bias. Standardized usability metrics, such as the System Usability Scale (SUS) [17] for VR comfort and the Igroup Presence Questionnaire (IPQ) [18] for immersion, could provide quantitative insights.

For practitioners, the key insight is to leverage VR's unique strengths - like personal immersion and scale - to enhance the conceptual message. Technical refinements should prioritize ergonomic hardware and intuitive spatial boundaries to maintain engagement. The medium shines when it actively reinforces the artistic intent rather than serving as mere decoration.



Mockup of a hybrid physical-VR installation, consisting of a minimalist box truss structure with four hanging blank canvases.

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